FM Part II, Lecture 2

Part II, Lecture 2: Agencies' Ratings, Internal Ratings, and How a Score Becomes a Probability of Default

Why this session matters

Lecture 1 ended with a promise: probabilities of default come from rating systems. This lecture opens the box. Three kinds of producers issue ratings, and they answer to different masters, look at different horizons and fail in different ways. Credit ratings inform investors and make bond issuance and syndicated loans possible, they feed credit risk quantification in risk management, they guide credit decisions, pricing and portfolio management, they drive the cost of capital for borrowers, and they sit inside banking regulation through the Basel framework. Understanding who produces a rating, and with what method, is what allows you to know how much to trust it.

Three producers, three philosophies

> **Agencies' ratings** come from S&P, Moody's and Fitch. They are typically Through-the-Cycle (TTC): built to measure risk across the whole economic cycle, so they move slowly and stay stable.

> **Internal ratings** are produced by banks for their own lending. They lean Point-in-Time (PIT): they reflect current conditions, including fresh behavioural data, so they move faster.

> **Market-implied ratings** are extracted from market data (credit spreads, credit default swap spreads, equity prices) transformed into a default-risk indicator. They are very Point-in-Time and can be volatile, because market sentiment and liquidity move daily.

Intuition for the TTC versus PIT split: a TTC rating asks "how would this borrower fare across a full cycle, including a recession?", so it does not jump at every quarter. A PIT measure asks "how likely is default in the next twelve months given everything we see right now?", so it reacts immediately, at the price of stability.

Agencies' ratings, in one paragraph of history and one of mechanics

The industry descends from 19th century mercantile credit agencies, through John Moody rating United States railroad bonds in 1909, to the 1941 merger that formed Standard & Poor's. Today's majors are S&P, Moody's and Fitch. Their ratings matter for capital markets (bonds, syndicated loans), with historical evidence of predictive power and methodologies codified over time. Agencies enjoy access to issuers' internal information; ratings can be solicited or unsolicited; and their only real asset is reputation, the thing that makes investors demand ratings and issuers pay for them. Under Basel II, agencies are recognised as External Credit Assessment Institutions for banks on the Standardized Approach.

Internal ratings and market-implied ratings

Internal ratings were developed from the mid-1990s and strongly boosted by Basel II's Internal Ratings-Based approaches. A bank uses them for its own lending decisions, pricing and capital calculation; they are confidential to the bank. They carry a higher Point-in-Time component than agencies' ratings because they exploit current conditions and behavioural data. They are natural for small and medium enterprise financing: proximity advantage, detailed information, and costs recovered through loan margins.

Market-implied ratings spread from the 1990s. The conversion of a spread or a price into a default-risk indicator relies heavily on assumptions that supplement the calculations, and the models are relatively simple. They reflect current market sentiment with low stability over time.

The matrix that organises everything

Ratings vary along two dimensions and two processes. This matrix is the backbone of the lecture:

	Assignment (ordinal output)	Quantification, calibration (cardinal output)
Issuer / borrower	issuer or borrower rating (the class, the letter)	ex ante probabilities of default, ex post default rates
Issue / facility	issue or facility rating	ex ante expected loss rates, ex post loss rates

Assignment ranks; quantification attaches numbers. Borrowers get probabilities of default; facilities get expected loss rates, because a facility's loss also depends on its own features (guarantees, seniority). Comparing ratings across different systems requires a master scale that maps each system's classes onto one common scale.

How agencies assign: the judgment-based approach, and why it can fail

Agencies primarily use judgment-based (principles-based) assignment. It is deliberately not mechanical: a rating is, in the end, an opinion. The process runs from preliminary analysis through meetings with the issuer, an analyst presentation, and a rating committee discussion and vote. The information range is deliberately wide: objective and subjective, quantitative and qualitative, historical and forward-looking, including sector analysis, competitive analysis and management quality. Financial ratios inform the committee but do not rigidly determine the rating. Afterwards, surveillance keeps going: ongoing monitoring, annual reviews, and triggers for action such as a CreditWatch placement, an outlook change, or a rating change.

Bhatia (2002) lists four possible causes of rating failure, balanced by one counterweight:

- **Information risk:** the data provided by the rated institutions are inaccurate.
- **Analytical constraints:** scarce human resources push agencies to lean on already issued rating reports instead of fresh analysis.
- **Revenue bias:** the issuer pays, and may be inclined to pay additional fees hoping for a more favorable rating.
- **Other incentive problems:** conflicts of interest from other services the agency sells, for example risk management advice.

The counterweight is reputation: the only asset that justifies investors demanding ratings and issuers paying for them. An agency that systematically sells good ratings destroys the very reason anyone buys them.

How banks assign: the statistical-based approach, step by step

Banks primarily use statistical-based assignment for internal ratings. Models interpret historical quantitative data: financial statements, internal behavioural data, credit bureau and credit register data. The output is a model-based rating, which for retail or simple segments is often the final rating, with no human override. These systems are increasingly enriched by new data (internet) and methods (machine learning, artificial intelligence), with regulators openly cautious about black boxes.

A typical architecture runs as separate modules per data source (financial scoring, behavioural scoring, credit register or bureau scoring, business sector rating, a questionnaire-based qualitative rating) feeding a model-based rating, then an override step that produces the borrower rating, and finally the facility rating with its loss-given-default rating. Modules earn their keep: easier maintenance, recalculation of a single source, a clearer customer profile, tolerance of missing data in one module, and room for specialized techniques per source.

The estimation itself builds what the course calls a reduced-form model:

1. Assemble a dataset of borrowers observed at a past “time zero”, including some who later defaulted and some who kept performing.
2. Candidate explanatory variables: only data that was available before the outcome was known.
3. Dependent variable: the outcome, coded 0 for performing and 1 for defaulted.
4. Techniques: stepwise selection to pick variables, then Linear Discriminant Analysis or logistic regression to set the weights.
5. Result: a score Z = a1 x X1 + a2 x X2 + … that best discriminates between defaults and non-defaults, meaning it minimizes both Type I errors (good borrowers rejected) and Type II errors (future defaulters accepted).

> **Worked example with two variables.** Suppose the selected variables are return on equity and return on assets, and estimation delivers Z = 200 x ROE + 300 x ROA, with a cut-off Z* = 30 (scores above 30 predicted to perform). Firm Alfa has return on equity of 8% and return on assets of 6%: its score is 200 x 0.08 + 300 x 0.06 = 16 + 18 = 34, above the cut-off, so it is predicted to perform. Firm Beta has return on equity of 5% and return on assets of 4%: its score is 10 + 12 = 22, below the cut-off, so it is predicted as a likely defaulter. Geometrically, the cut-off defines the separating line ROE = (Z*/a) - (b/a) x ROA, here ROE = 0.15 - 1.5 x ROA. Note what the equation says: a low return on equity is acceptable if the return on assets is high enough. That interaction is exactly what makes the model multivariate rather than a pair of independent thresholds.

> **From one cut-off to rating bands.** A binary accept/reject needs a single cut-off. Assigning an ordinal rating class needs several: Z1, *Z2*, Z3* and so on. Each new cut-off is a line with the same slope and a different intercept, so the bands are parallel. With the model above, cut-offs at 30, 34 and 38 would create four bands. Higher score means higher predicted quality and a lower expected default rate, and each band is then calibrated to its own probability of default through the long-run average default rate seen in Lecture 1.

The balance between automated and labor-intensive techniques varies with the borrower’s complexity and the loan’s gross contribution: statistical algorithms such as discriminant scoring handle installment loans and small business, while human-based procedures cover the qualitative aspects of large corporate borrowers that cannot be objectively measured.

Time-frame decisions the regulators care about

- The common observation period for PD estimation is one year. Longer horizons make models more forward-looking but reduce predictive power, and behavioural data lose relevance as the horizon lengthens.
- Basel II (paragraph 414) and the ECB Guide (2024) expect rating assignment to target a longer horizon, two to three years according to the ECB, while PD quantification stays at one year. Assignment looks far, calibration looks near.
- The link between time-zero data and observation-period outcome must be consistent for every borrower in the dataset.

- Time-stratification, using different time zeros for different borrowers, reduces dependence on a single calendar year or one stage of the economic cycle, smoothing the model's relation to the cycle.

The assigned reading: how accurate are ratings, measured with the Gini coefficient

The Lecture 1 assignment is an extract of S&P's 2024 Annual Global Corporate Default and Rating Transition Study (Appendix II). It measures rating performance with the Lorenz curve and the Gini coefficient. The Lorenz curve plots the cumulative share of issuers on the horizontal axis against the cumulative share of defaulters on the vertical axis, with both ordered from the lowest rating ('CCC'/'C') to the highest ('AAA').

> **Reading a point on the curve.** If 'CCC'/'C' rated entities make up 10% of all issuers at the start of the period and account for 50% of the defaulters, the point (10, 50) sits on the curve. If ratings approximated default risk only randomly, the curve would fall along the diagonal and the Gini coefficient would be zero. If ratings ranked risk perfectly, so that every default occurred among the lowest-rated entities, the curve would capture all the area above the diagonal and the Gini coefficient would be one. The coefficient itself is area B divided by area A plus area B, where B is the area between the actual curve and the random diagonal and A is the area between the actual curve and the ideal curve.

The study's Table 27 reports corporate Gini coefficients by region over 1981-2024: the one-year weighted average is 82.86% (standard deviation 5.33), against 75.11% at three years, 71.72% at five years and 69.22% at seven years. Two facts to retain: accuracy is high in absolute terms, and it decays as the horizon lengthens, exactly the trade-off the time-frame section predicts.

What you must walk away with

Must know:
• Three producers: agencies (typically Through-the-Cycle), banks' internal ratings (Point-in-Time leaning, boosted by the Internal Ratings-Based approaches, confidential), and market-implied ratings (very Point-in-Time, assumption-heavy, volatile).
• The matrix: issuer versus issue on one side, assignment (ordinal) versus quantification (cardinal) on the other; borrowers map to probabilities of default, facilities to expected loss rates.
• Bhatia's four causes of rating failure: information risk, analytical constraints, revenue bias, incentive conflicts; reputation is the counterweight.
• Statistical models are modular, produce a reduced-form score Z = sum of ai x Xi, minimize Type I and Type II errors, and turn one cut-off into parallel rating bands via multiple cut-offs.
• Assignment targets a two-to-three-year horizon (Basel II paragraph 414, ECB 2024), PD quantification stays at one year, and time-stratification smooths cycle dependence.
• Rating accuracy is measured with the Lorenz curve and the Gini coefficient: zero means random, one means perfect ranking; S&P's one-year corporate average over 1981-2024 is about 83% and decays with the horizon.

Reading a credit classification matrix

A model can make two different mistakes: accept a borrower that later defaults, or reject a borrower that would have repaid. Read the actual-outcome column first, then the predicted-outcome row. The denominator is the actual class, not all borrowers accepted by the bank.

Original illustration. In a balanced sample of 100 future defaulters and 100 non-defaulters, a model accepts 8 future defaulters and rejects 4 non-defaulters. It identifies 92% of future defaulters and correctly accepts 96% of non-defaulters. Overall accuracy in this balanced sample is (92 + 96) / 200 = 94%.

The share of accepted borrowers who default is a different question: 8 / (8 + 96) = 7.69% in this sample. It cannot be read directly from the 8% error rate. Nor is 7.69% a calibrated population PD: a deliberately balanced sample has a default frequency unlike most lending populations.

A longer prediction horizon often makes classification harder. Compare errors at one and two years using the same actual classes and definitions. A rise in missed defaults means deterioration in default detection, even if rejection of good borrowers remains rare. Do not average the two horizons as if they were independent observations.

An in-sample result describes the firms used to fit the model. It is not evidence of equal performance on new borrowers. Ask for an independent test sample, later-period validation, the population default frequency, and the cost of each error. A missed default and a rejected sound borrower do not impose the same economic cost.

Must know. State the actual class, the prediction error, the denominator and the horizon. Distinguish class-specific error rates, overall accuracy and default frequency among accepted borrowers. Use descriptive error names because Type I / Type II labels depend on convention. Good classification does not by itself establish PD calibration or future out-of-sample performance.